

Generalizable Brain Tumor Segmentation with Self-Training and Tumor-Aware Deformations

Henrique Zan Grande¹ · Jeovane Honorio Alves²
Rayson Laroça¹ · Andre Gustavo Hochuli¹

¹ Pontifical Catholic University of Paraná (PUCPR), Brazil

² University of Luxembourg, SEDAN / SnT, Luxembourg

15th BraTS Cluster of Challenges @ MICCAI 2026 | BraTS-GoAT 2026 – Generalizability Across Tumors

PROBLEM AND MOTIVATION

A segmentation model is only useful if it holds up on tumors, scanners and populations it never saw in training.

- Problem:** Segmentation models such as nnU-Net lose accuracy when tumor type, acquisition protocol or patient population shifts from the training distribution.
- Motivation:** BraTS-GoAT evaluates that shift across gliomas, meningiomas, metastases, African and pediatric cohorts, the heterogeneity a clinical deployment actually faces.

RQ1

Does self-training on the unlabeled data improve generalization?

RQ2

Does tumor-aware augmentation provide gains beyond those of self-training?

DATASET

1,351

labeled volumes

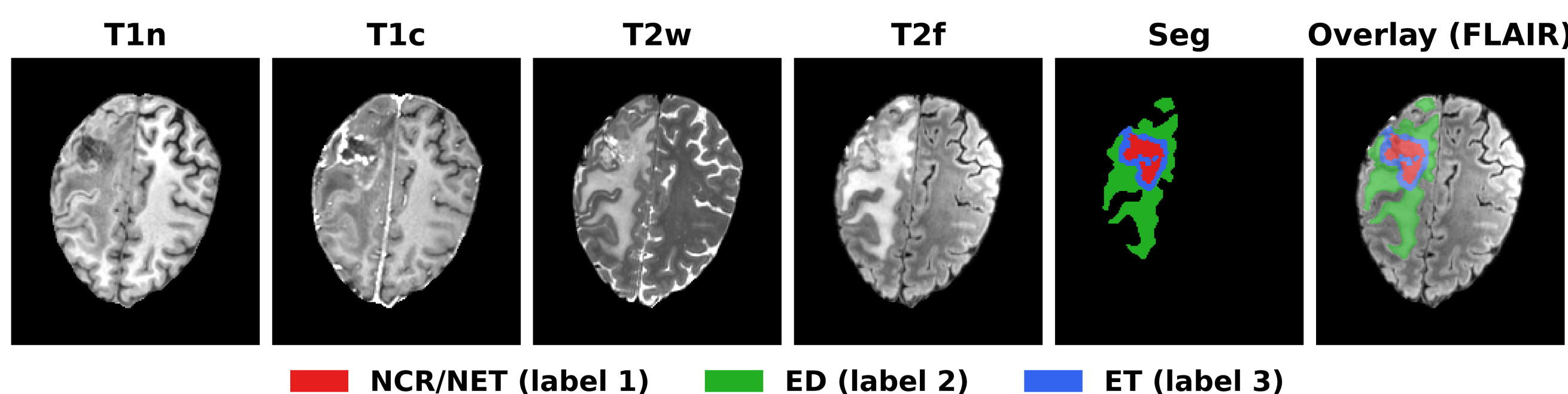
1,138

unlabeled volumes

451

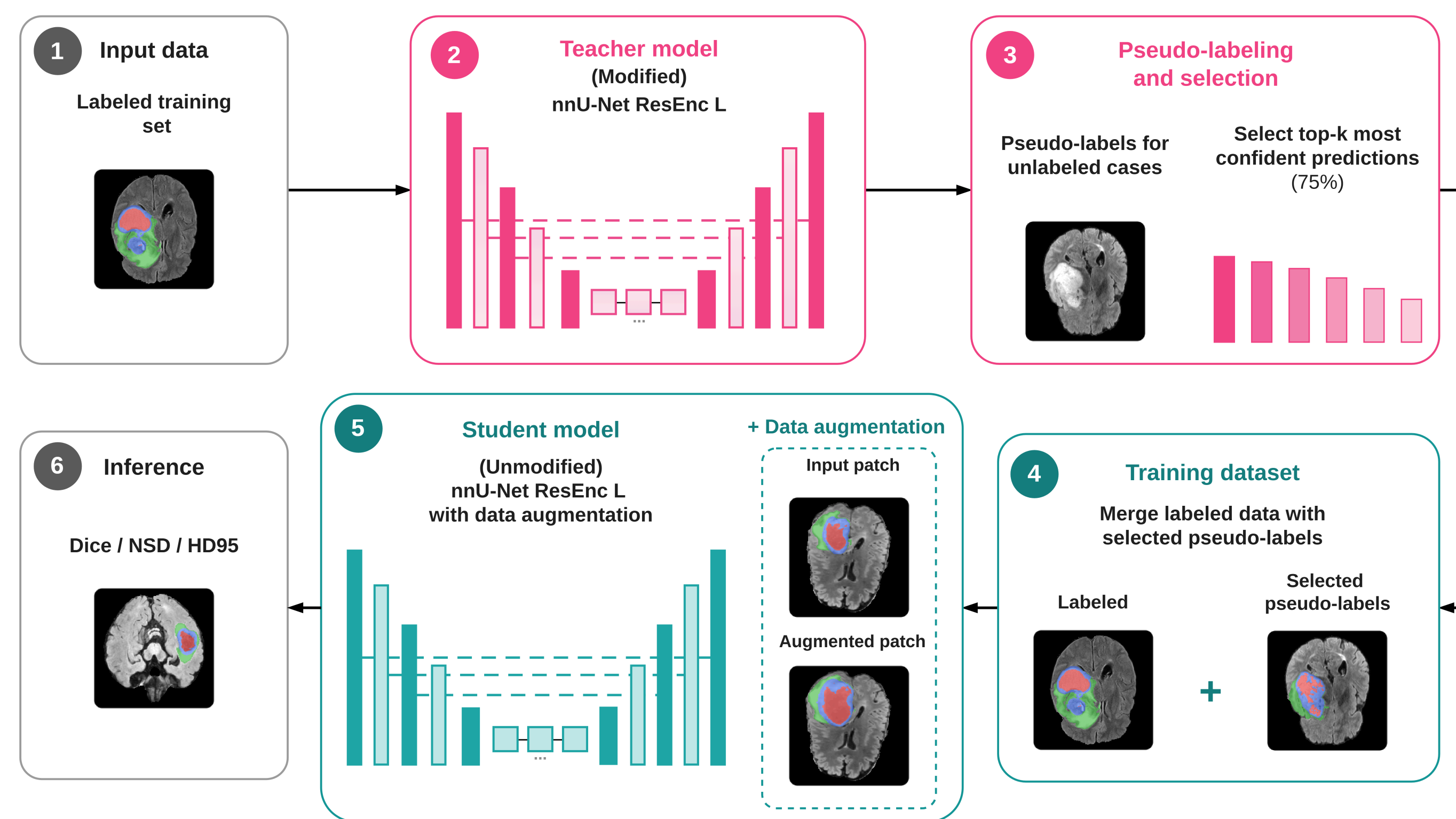
validation volumes

- Training:** The full BraTS adult-glioma training set, labeled, plus meningioma and brain-metastasis data that is labeled for a subset only and unlabeled for the rest.
- Validation:** Adult gliomas, BraTS-Africa, meningiomas, brain metastases and pediatric tumors: five cohorts, two unseen in training, four MRI modalities per case.

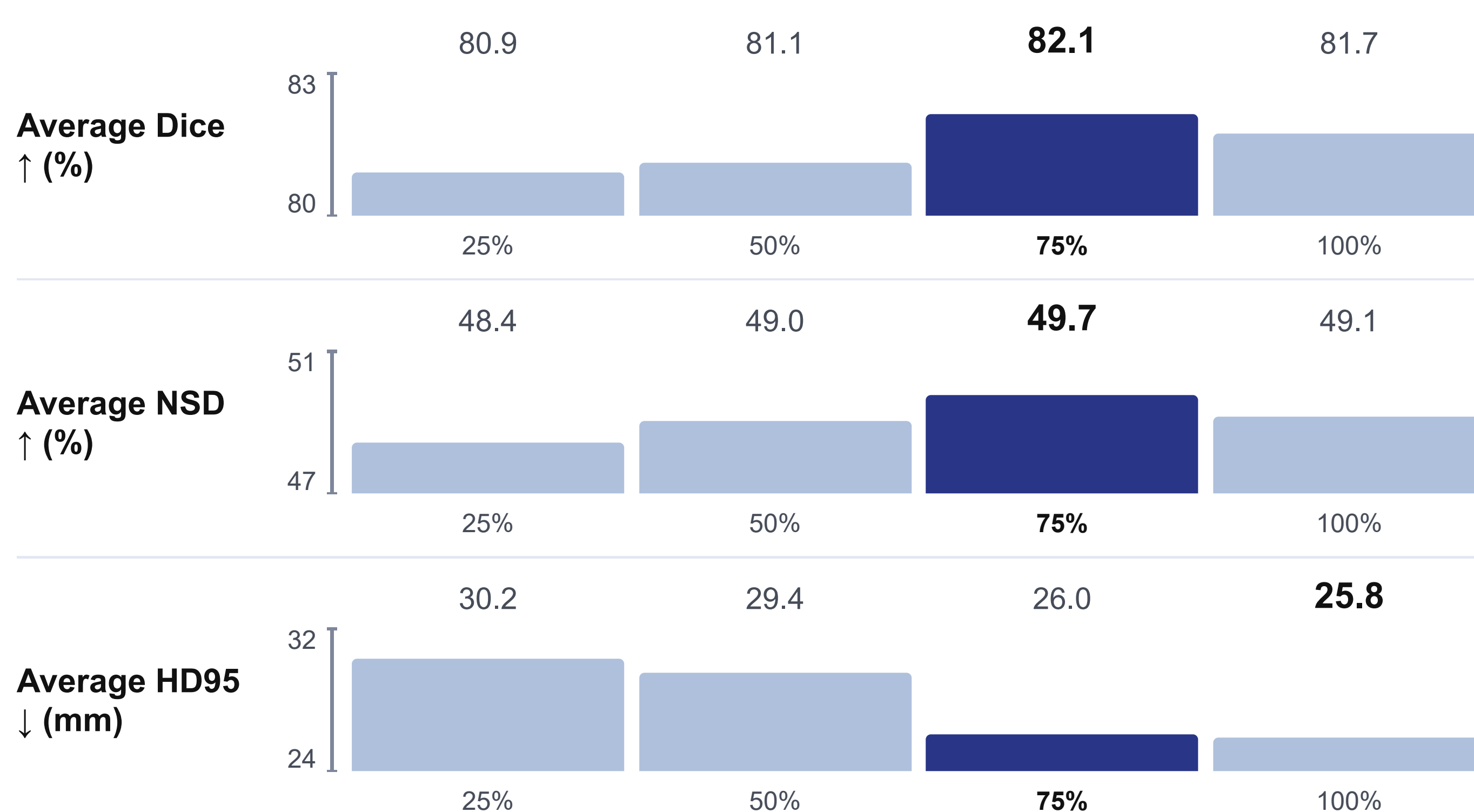


METHOD: TEACHER / STUDENT SELF-TRAINING

A high-capacity teacher writes pseudo-labels; a standard nnU-Net student learns from labeled plus selected cases.



How much of the pseudo-labeled data to keep

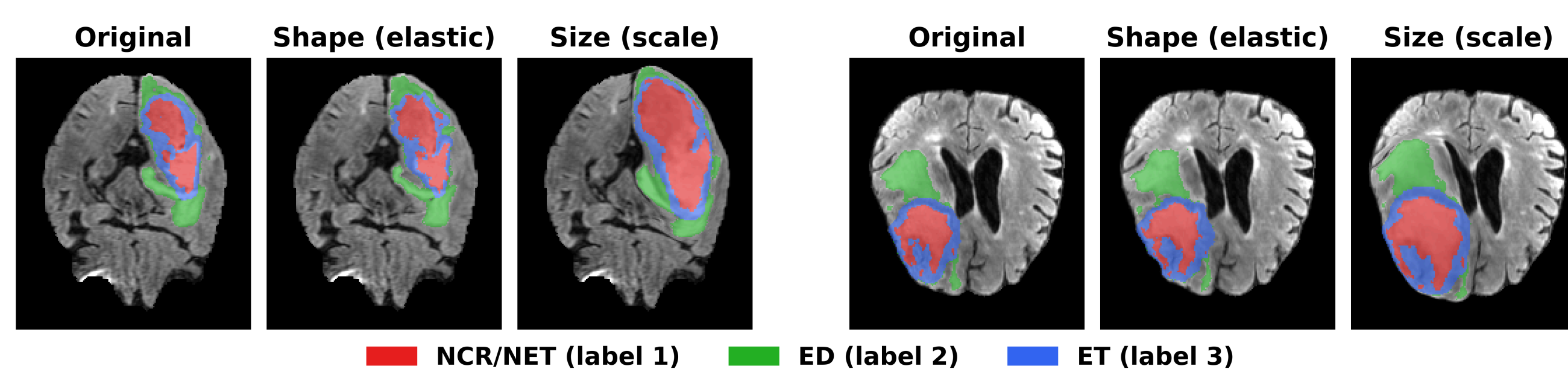


- Retaining the 75% most confident pseudo-labels ($\geq 97.22\%$, 803 of 1,071 non-empty cases) yields the highest Dice and NSD; the 0.2 mm lost on HD95 is offset by +0.4 Dice and +0.6 NSD.

METHOD: TUMOR-AWARE DEFORMABLE AUGMENTATION

Reshape the lesion, leave the surrounding anatomy untouched.

- Elastic deformation or isotropic scaling about the tumor center of mass, applied with equal probability; tiny lesions are excluded.
- Confined to the lesion and a narrow halo by a Gaussian-smoothed map from the dilated mask. Image and labels share one displacement field.



RESULTS

82.4

avg Dice ↑ (%)

49.9

avg NSD ↑ (%)

25.4

avg HD95 ↓ (mm)

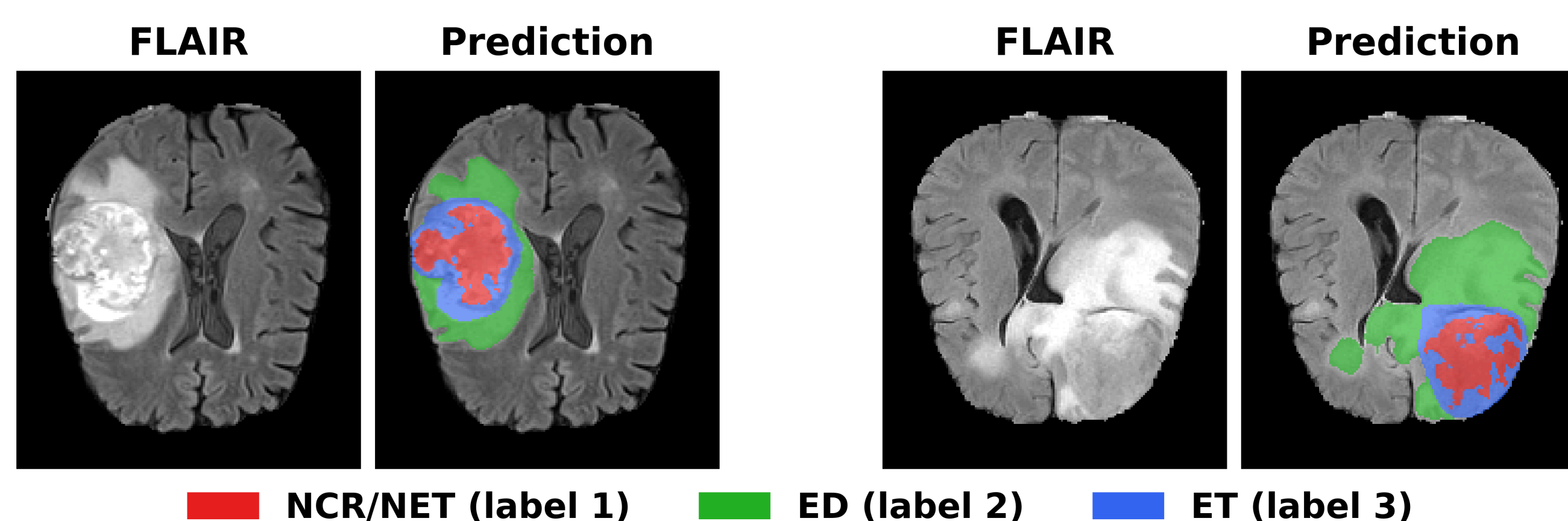
All models on the BraTS-GoAT validation set

Model	Metric	WT	TC	ET	Avg
Teacher	Dice ↑	86.7	80.2	77.3	81.4
	NSD ↑	45.5	48.3	53.0	48.9
	HD95 ↓	16.9	25.0	41.0	27.6
Student, labeled only	Dice ↑	86.6	81.1	76.0	81.2
	NSD ↑	46.8	48.6	52.1	49.1
	HD95 ↓	19.3	25.6	46.8	30.6
Pseudo-labels	Dice ↑	87.8	80.3	77.5	81.9
	NSD ↑	46.9	48.4	53.4	49.5
	HD95 ↓	15.6	23.3	39.2	26.0
Augmentation	Dice ↑	87.3	81.5	76.9	81.9
	NSD ↑	47.3	49.9	53.5	50.2
	HD95 ↓	15.2	18.7	41.9	25.3
Final	Dice ↑	88.1	81.7	77.5	82.4
	NSD ↑	47.3	49.0	53.3	49.9
	HD95 ↓	14.0	21.5	40.7	25.4

- WT** whole tumor · **TC** tumor core · **ET** enhancing tumor · **Dice** Dice similarity coefficient (%) · **NSD** normalized surface Dice (%) · **HD95** 95th-percentile Hausdorff distance (mm).

- The final model leads on Dice by 0.5 (82.4 vs 81.9); augmentation-only stays marginally ahead on NSD by 0.3 and HD95 by 0.1 mm. Both clearly outperform the labeled-only student.

Qualitative results on validation cases



CONCLUSIONS

Self-training and tumor-aware deformable augmentation lift average Dice from 81.2 to 82.4 and NSD from 49.1 to 49.9, while average HD95 falls from 30.6 mm to 25.4 mm.

- The two components are complementary:** Pseudo-labels broaden the training distribution and the augmentation adds shape robustness, with the largest gains on WT and TC.
- Confidence filtering matters more than volume:** Retaining 75% of the pseudo-labels outperforms retaining all of them.
- The teacher offers little direct advantage over the baseline student despite being more complex; its contribution lies in generating reliable pseudo-labels.
- Next steps:** Iterative self-training, teacher ensembles, richer ranking criteria and per-domain analysis across modalities and tumor types.

